

Automated Robotic Soccer (Robot vs. Robot)



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Final Year Project Report

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Fall 2020

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Dedication

We would like to dedicate this project to our parents, for their strong and steady support and encouragement throughout our academic progress. We would also like to dedicate this project to our advisors and instructors for their guidance that have directed our path. And also to our friends for their encouragement and companionship.



Department of Computer Engineering
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Automated Robotic Soccer (Robot vs Robot)

Presented by

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In partial fulfilment of the four years **B.S. Computer Engineering** requirements, HITEC University Taxila, Pakistan underneath the supervision of its project adviser as well as authorized by the project review board has been recognized.

Dr. Raza Ali Shah
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Abstract

This research project focuses on the development of a real-time planning and execution system for robots, using robotic soccer as a practical testing ground. Multiple prototypes were produced with initial prototype using MPU6050 for localization, but it has a very high error so we needed to shift to image processing-based using OpenCV and an android camera. This system was successful in robot localization and effective real-time communication. This research aligns with the United Nations Sustainable Development Goal 9, promoting industry, innovation, and infrastructure.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
IMU	Inertial Measurement Unit
FDA	Food and Drug Administration
RISC	Reduced Instruction Set Computing
Kbytes	Kilobytes
MPU	Motion Processing Unit
ESP	Espressif Systems (refers to the ESP series of microcontrollers)
WiFi	Wireless Fidelity
DMP	Digital Motion Processor
FOV	Field of View
IP	Internet Protocol
HSV	Hue, Saturation, Value
SDGs	Sustainable Development Goals
STM	STMicroelectronics
BJT	Bipolar Junction Transistor
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
GPS	Global Positioning System
GNU	GNU's not Unix
GPU	Graphical Processing Unit
CPU	Central Processing Unit

Chapter One

1 INTRODUCTION

1.1 Overview

Robotics in the modern era has seen significant advancements, leading to the development of autonomous systems capable of performing complex tasks with minimal human intervention. The ability of accurate localization, navigation, and execution within dynamic environments is very important for these systems to function effectively.

The growing trend of robots usage worldwide necessitates the need to find optimal solutions to control them. Development of a robot system capable of operating dynamically can be a tough and challenging task. To develop and test new algorithms for these robots, Robotic Soccer can be a practical and realistic testing ground because of its real-world and dynamic nature, its requirement of the usage of multiple sensors and the processing of that data to plan appropriate actions. This can help with localization of robots, communication between robots and object detection. The knowledge gained through these experiments can be valuable in designing systems such as robotic automation systems, group transport systems and search and rescue missions etc.

1.2 Problem Statement

Developing a real-time planning and execution system for robots requires the complex task of designing and implementing robots capable of operating in a dynamic environment which is a challenging task. The robots must collaborate with other robots nearby optimizing their actions to accomplish a task efficiently. This challenge requires effective real-time communication, precise robot localization and navigation, on-the-edge object detection and optimal pathfinding, making it a task with highly complex real-time processing.

In our project, we aimed to develop a system that allows multiple robots to operate autonomously without any human interaction on a predefined playing field.

1.3 Goals and Objectives

Our project has the following main objectives:

- Advancement and Innovation in Robotics
- Benchmarking and Evaluation of Robots in a controlled environment

1.4 Sustainable Development Goals

Our project targets the 9th goal of the UN Sustainable Development Goals that is Industry, Innovation and Infrastructure. Our project aims to optimize the performance of robots in industry and make robots perform better in ever changing environment.

By optimizing the performance of robots in dynamic and ever-changing environments, our project adds to the advancements in industrial automation and innovation. Accurate robot localization, efficient real-time communication, and navigation capabilities are critical for improving the productivity and adaptability of robots in industrial settings. Our system aims to improve the operational efficiency and effectiveness of robots.

1.5 Why Robotic Soccer?

The reasons to choose Robotic Soccer over other projects are as follows:

- It is a real-world task that involves complex actions carried out with precision and speed.
- It imposes real time constraints on the embedded system which must respond appropriately to changes in the configuration of the play field. This forces the system to process incoming data and decide on suitable actions within a short time interval.
- It requires the use of multiple sensors and the processing of that data to plan appropriate actions.
- It requires integration of all the concepts we have studied thus far i.e. code optimization, sensor and noise management, building reliable software, responding in real-time to sensor input, and planning and new concept that we will learn such as robot localization.

This project provides us with an opportunity to bring together all the tools we have acquired during the term, strengthen and improve our understanding of course concepts, and apply the skills we have been developing over the past 3 years toward solving a challenging problem.

1.6 Accessing the source code

The source code for the project is hosted on Codeberg and is publically available to look at and modify at the following link: [Automated Robotic Soccer](#). The complete source code is uploaded and is liscensed under GNU General Public License version 3.

Chapter Two

2 LITERATURE SURVEY

2.1 Overview

In the period from 1990 to 2024, significant milestones in robotics were achieved. In 1994, the FDA cleared the Cyberknife, a groundbreaking robot-assisted surgery system. The 2000s brought consumer robots like the Roomba and advancements in AI and sensors. In 2005, the Mars rovers Spirit and Opportunity landed on Mars, significantly surpassing their expected lifespans. The 2010s witnessed the rise of humanoid robots, improvements in AI, and the proliferation of specialized non-humanoid robots. Moore's Law and open-source software enabled greater computational power and AI integration in robots. Self-driving cars emerged, with Tesla leading in autonomous capabilities. However, concerns grew about data privacy and ethics, particularly regarding the rights and responsibilities of robots. The decade also saw the development of nanobots for medical applications, with the potential to revolutionize healthcare.

Between all of this, RoboCup was proposed by a group of researchers from various institutions (including Hiroaki Kitano, Minoru Asada, Yasou Kuniyoshi) in 1996 (Kitano, Asada, Kuniyoshi, Noda, Osawa 1996, 1). The long term goal of RoboCup is to develop a robot soccer team which beats the world champion within the middle of the next century . The first RoboCup competition was held 1997 in Nagoya, Japan and the subgoal was “built a real robot soccer team which can play reasonably well with modified rules”.

2.2 The Robocup Initiative

The RoboCup initiative aims to advance artificial intelligence (AI) and robotics through the challenge of robotic soccer. The initiative is designed to support research and development in AI, robotics, and related fields by providing a common testing ground that is interesting, practical, and challenging.

The primary goal of RoboCup is to promote research and education in the fields of AI and robotics. The soccer game unlike chess which was used to train AI's provides challenges including real-time decision making, multi-agent collaboration, and sensor-based control through its dynamic and on the spot execution.

The RoboCup competition is broken into several leagues, each designed to address different aspects of robotics and AI. These include the Simulation League, the Small Size Robot League, the Middle Size Robot League, and the Humanoid Robot League, among others.

RoboCup also has a significant educational component, encouraging students and researchers to engage in hands-on learning and collaborative problem-solving. It serves as a platform for knowledge exchange and innovation.

The document[1] outlines future goals, including making robots capable to eventually compete against human teams. This target serves as a long-term vision to drive ongoing research and development in the field.

2.3 10 Years of Achievements and Future Challenges of RoboCup

The RoboCup Federation aims to develop autonomous humanoid robots capable of competing against human soccer players by 2050. RoboCup competitions have significantly advanced robotics and AI.

This document[2] provides an overview of the RoboCup initiative, reflecting on a decade of progress and future directions. The 2006 World Championship in Bremen, Germany, showcased notable progress. The competitions have supported innovations in various fields. While chess has been a traditional measure of AI, soccer requires handling complex, real-time interactions, making it a more comprehensive test of everyday intelligence.

Developing robots that can understand and interact with their environment in real-time remains a key challenge. The integration of body and intelligence in robotics is crucial for success in RoboCup. Robots must act appropriately with humans, ensuring that their performance is effective and harmless.

Chapter Three

3 METHODOLOGY

In order to implement the product, we had to write software to handle a dynamic environment, with an opposing agent whose behavior cannot be predicted in advance. State-based AI's would have been used to provide robots with a suitable set of behaviors to handle specific tasks. The structure of the robots will be divided into multiple modules, responsible to accomplish a specific task assigned to them.

Localization of robots was the first target of the project. Our solution was to divide the field into multiple pixels using a coordinate system. All robots will calculate their positions on the coordinate system and communicate it to their teammates. Similarly, robots will detect their opponent players and the ball and make their decision by communicating the situation at hand to their teammates.

3.1 Funtional Requirements

Our project needed to implement the following functional requirements:

- Localization of robots on the 2d plane field.
- Detecting the ball and calculating its distance.
- Communication and synchronization between same team robots
- Detecting the opponent goal post and getting the ball through it

3.2 Localization of robots and the ball

The first target of our project was to have accurate Localization of the robots. Localization in robotics is the process of determining exactly where a certain object is located at a certain time with respect to its environment. It is the first step to deal with while doing any project regarding autonomous cars because a car would be able to carry out a task in a target environment only if it know where it is located in the said environment.

Lets suppose we have an object residing in an x-y plane and we have to move it from the point (x,y) to (x_1, y_1) it first need to know where x , x_1 , y and y_1 , only then can it move to the desired location like shown in Figure 3.1.

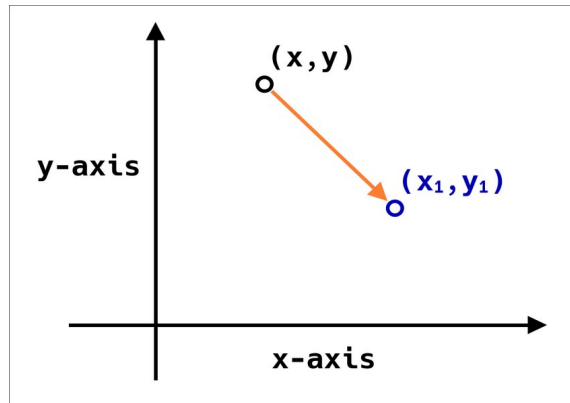


Figure 3.1: Depiction of an object at coordinates (x,y) moving to (x_1, y_1)

3.2.1 Initial approach to localization

The field was divided into several pixels through a coordinate system. Each robot started at a fixed position and moved calculated its current position at point in the game relative to that starting position. All the the robots used the same reference point and its position had to be calculated accordingly. An inertial measurement unit (IMU) was decided upon to be used as the localizer.

3.2.1.1 Coordinate System

The playing field was measured 4 meters in length and 3 meters in width and was divided by us into a grid of pixels, such that each pixel had a specific coordinate (x, y). This coordinate system acted as the common frame of reference for all robots. Each robot started at a predetermined starting position with known coordinates within the grid.

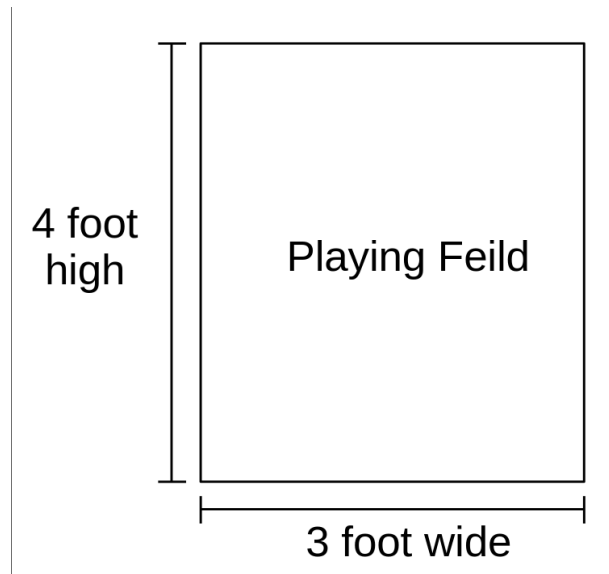


Figure 3.2: Playing field dimensions

3.2.1.2 Position Calculation

Throughout the game, each robot continuously had to calculate its current position relative to its starting position. This position was updated at real-time and dynamically. An Inertial Measurement Unit (IMU) was used for localization. The IMU was used to get data on the robot's acceleration, velocity, and orientation, which was then used to estimate the robot's current location and its movement from the starting point.

3.2.1.3 Inertial Measurement Unit (IMU)

An IMU is a device which measures and reports a body's specific force, angular rate, and sometimes the orientation, with the help of a combination of accelerometers, gyroscopes, and magnetometers. Each robot used an IMU to provide continuous updates on its movement. By processing the data from the IMU, we could calculate the robot's displacement from its starting point.

3.2.1.4 Localization of the ball

The localization of the ball was also done using similar principles as the robots. Its position was calculated relative to a fixed reference point on the field. Another IMU inside of the ball was used to find its speed and its direction of movement.

3.2.2 Challenge with initial approach

Sensors generally are very prone to noise and errors especially when moving constantly and the IMU we used was no exception. It had a really high error percision error with a drift of about 6 meters per minute. This error was problematic as our field itself was only 3 foot in width, the error per minute was several times the width of our field. As, both robots and ball localization systems were based on localization by the same IMU, this was a big issue.

3.2.2.1 Impact on Localization Systems

The reliance of both the robots and the ball's localization systems on the IMUs, made the high error rate a critical issue. The inaccuracies and drift caused unreliable position estimates, which undermined the entire localization framework.

3.2.3 Transition to Image Processing Approach

Image processing was already in our idea bank as a backup approach if our initial approach fails. Given the limitations and challenges of the IMU-based approach, we decided that it was time to try it and shift to an image processing-based model using a mobile camera for localization.

The new approach utilizes a mobile camera to track and localize both the robots and the ball. This method relies on image processing techniques to extract positional information from the visual data captured by the camera. This was already in our idea bank as a backup approach if our initial approach fails. Given the limitations and challenges of the IMU-based approach, we decided that it was time to try it and shift to an image processing-based model using a mobile camera for localization.

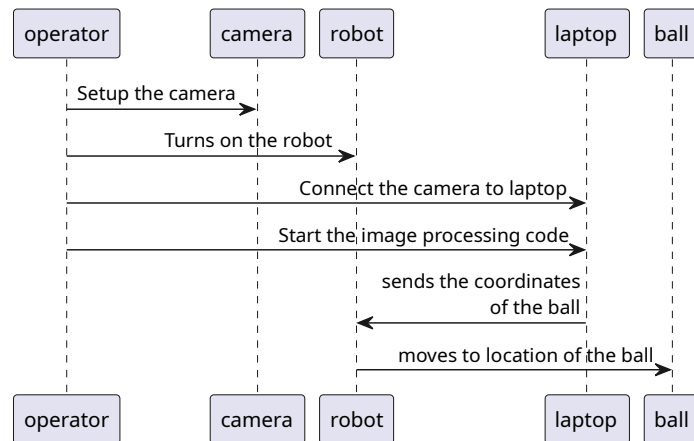


Figure 3.3: Sequence diagram for the project

3.2.3.1 Field Preparation

To optimize object detection and tracking, the playing field was prepared accordingly. The field was painted white to enhance contrast, thus facilitating the detection of objects placed on it. Objects were also given different colors i.e. the ball was colored dark green to distinguish it from the white background and each robot was had colored paper glued on their roofs so that one robot had a red roof while the other had a purple roof.

3.2.3.2 Object Localization using Image Processing

Image processing algorithms were used to differentiate objects based on their colors. Color segmentation helped with the localization of robots and the ball on the white background.

The localization was done using the following methodology. The project used a live video feed captured by a mobile camera, thus providing real-time footage of the playing field. The

frames from the video feed were converted to the HSV (Hue, Saturation, Value) color space. Which is useful for color-based segmentation as it separates color information from brightness information, making it easier to identify objects based on their color. The HSV frames were thresholded using predefined lower and upper bounds for specific colors. This resulted in binary masks highlighting regions of interest corresponding to the desired colors.

Contours were then identified within the binary masks which represented potential objects of interest, such as the ball and the robots. For each detected contour, bounding boxes were drawn around the corresponding objects on the frame. These bounding boxes were used as visual cues for the positions and were crucial for finding dimensions of the objects. The center points of the detected objects were calculated to determine their precise coordinates on the playing field.

There were also two additional markers placed on the roofs of each robot to find their orientations. By analyzing the arrangement of these markers relative to the camera's axis, the angle of each robot with respect to the camera got computed.

Now, the coordinates and other relevant information, such as the radius of the detected ball, were packaged into a byte array for transmission from the image processing system to the microcontrollers within the robots.

The activity diagram for the object detection is shown below:

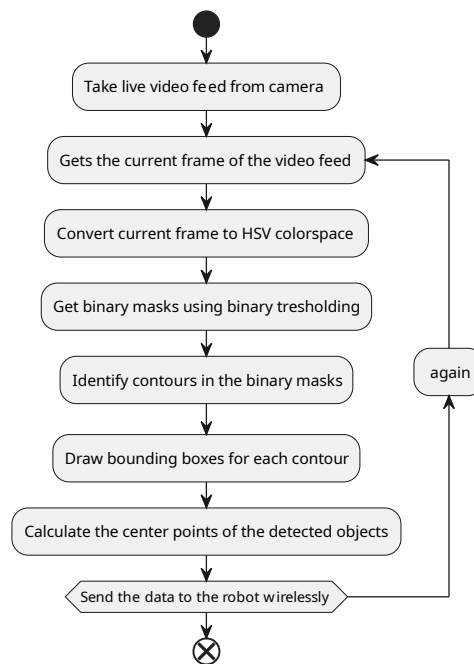


Figure 3.4: Activity state diagram for the project

Chapter Four

4 SYSTEM DESIGN AND IMPLEMENTATION

4.1 Overview

This chapter consists of the practical system design based off the methodology of our final year project. Throughout the span of our fyp, we went through multiple newer system designs due to some difficulties we faced in the older designs.

We started with an STM32 based system with the MPU6050 IMU using the ESP-01 wifi module but then we shifted to ESP32 due to our over-reliance on Wifi and ESP32 having builtin wifi. But due to the failure of MPU6050, we decided to move towards image processing in Python with opencv for localization and an android phone camera for the live feed.

As we have had multiple design of our system, we will be going in detail of every single one of them now.

4.2 STM32 based design

This design of ours was the one we proposed in our pitch presentation. It used a STM32F401 Microcontroller as STM32 devices are known for their high performance while being both low power and cost efficient. STM32F401 features a ARM® Cortex® -M4 32- bit RISC core operating at a frequency of up to 84 MHz. It also includes 128 to 512 Kbytes of Flash, 96 Kbytes of SRAM and the available packages range from 49 to 100 pins. We did have to switch to a similar microcontroller STM32F103 due to its better availability in Pakistan.

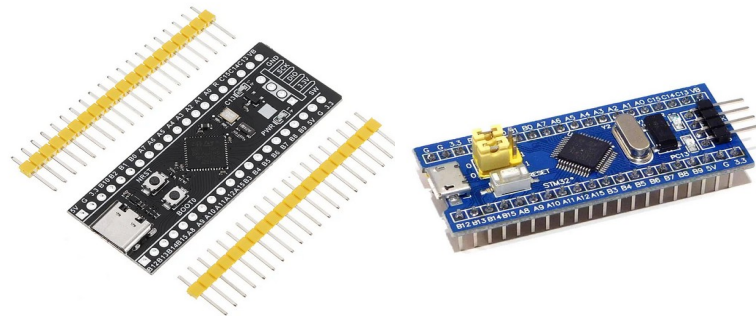


Figure 4.1: STM32F401 (left) and STM32F103 (right) microcontrollers

This design of ours also used MPU6050 Motion Processor which offers a offering a compact 6-axis sensor combining a 3-axis gyroscope, 3-axis accelerometer, and a Digital Motion Processor (DMP) in a small 4x4x0.9mm package. It provides precise motion tracking eliminates the need for costly and complex integration of discrete components, making it ideal for applications like smartphone gesture commands, gaming, augmented reality, and navigation.

The STM32 boards would have been programmed using the C programming language because of support of vast majority of embedded devices with C compilers.

Stlink would be used to burn the code onto the board. Stlink is an open source toolset to

program and debug STM32 devices and boards manufactured by STMicroelectronics. It consists of tools like st-info (a programmer and chip information tool) and st-flash (to burn the code on the controller).

4.3 First prototype of the robot

The main microcontroller of the project was soon changed to ESP32 because of its builtin wifi capabilities and sue to the ESP-NOW protocol only supporting esp devices. As the communication between our devices was planned to be peer to peer, ESP-NOW was a huge help in that. It was decided to program the esp decices using Arduino IDE 2.0 because of its huge open-source support. The esp-idf programming kit was used to program esp32 through arduino ide.

The first prototype was a two story robot car utilizing three 3.7 volt batteries. One of the batteries was used to power the esp32 while the other two with a combined voltage of 7.4 volts was used to power the motors required to move the robot. MPU6050 was used to find the orientations of the robot.

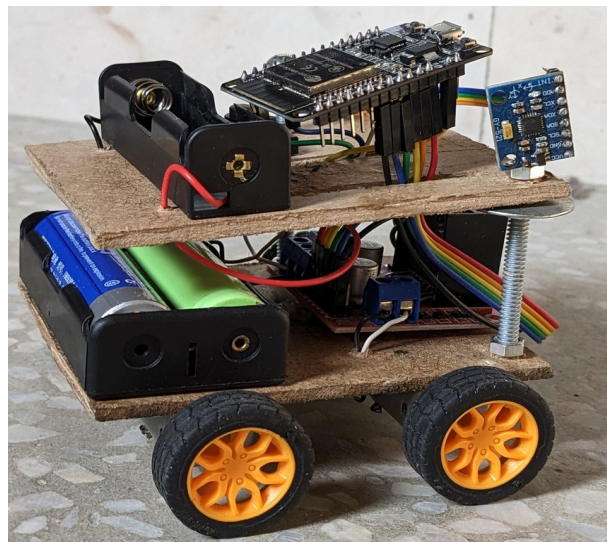


Figure 4.2: The first prototype robot

L298N Motor Driver Module is a high power motor driver module for driving DC and Stepper Motors. This module consists of an L298 motor driver IC and a 78M05 5V regulator. L298N Module is capable of controlling up to 4 DC motors, or 2 DC motors with directional and speed control.

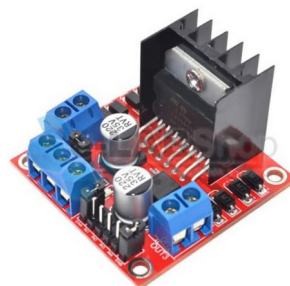


Figure 4.3: The L298N motor driver

The motors used in the prototype were small local dc motors with high rpm. Small tires were used along with the motors.

Figure 4.3 shows the circuit diagram for the prototype depicting all the circuit parts and their connections.

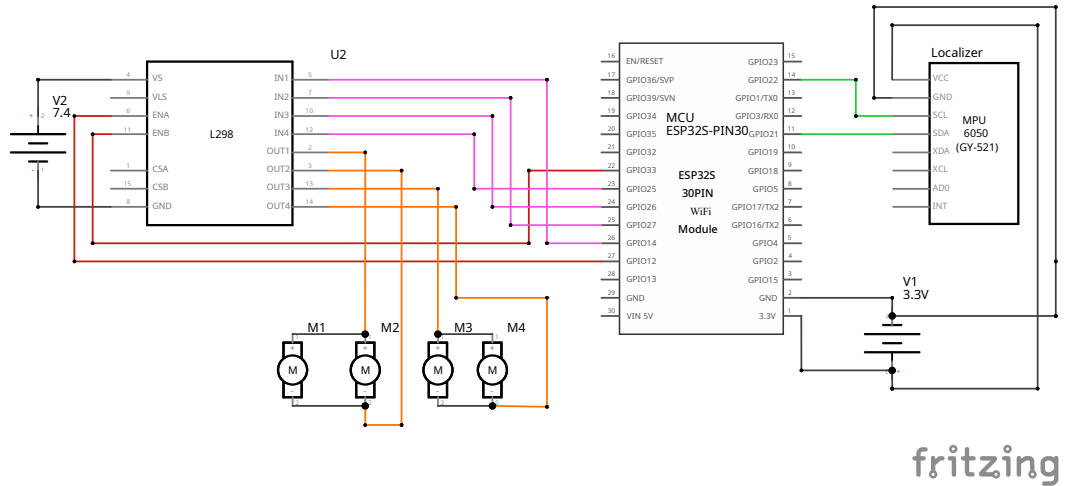


Figure 4.4: Circuit diagram for the first prototype robot

4.4 Second prototype of the robot

The first prototype had some noticable issues with it which had to be addressed. The first one was the use of L298N motor driver. We did use the motor driver in an older project thus it was our initial choice for this project too but the L298N is quite an old chip which uses Bipolar junction transistors (BJTs) for its H-bridge and have a noticable voltage drop of about 2.3 to 2.7 volts. The L298N also has a big overheating problem which is again excessive voltage getting lost. In order to deal with this issue, we switched to the TB6612FNG motor driver.

TB6612FNG motor driver is also an H-Bridge driver that can turn a connected motor in either direction and at a variable speed. The main benefit of the TB6612FNG is its use of MOSFETs.

A MOSFET is a metal oxide semiconductor field-effect transistor having an extremely low, almost negligible, voltage drop. This means that nearly all the voltage from the power supply is delivered to the motor. And, with very little voltage drop to deal with, there is very little heat to dissipate. Along with that, a TB6612FNG also comes with built-in thermal shutdown and a standby power mode for better battery life. The smaller size of TB6612FNG as compared to the L298N also helped us fit all the circuit of the robot on a single platform without a need of a two story robot.

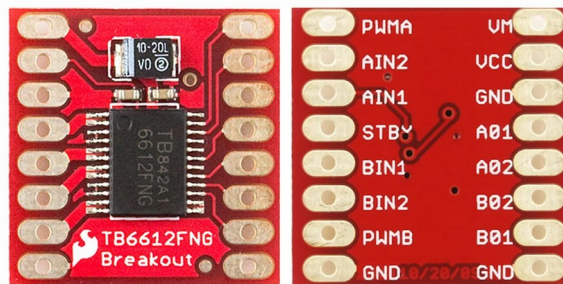


Figure 4.4: The TB1612FNG motor driver

The second issue with the first approach was the use of local high rpm motors that were unable to move the car with even a small weight. These motors were replaced with high torque gear motors of small size to fit in with the robot size. The tires were also changed accordingly.

Another big and important issue was that we gave the ESP32 an input voltage of 4.2 volts when the maximum it can handle is 3.3 volts. To solve this we use a X16009 voltage booster to boost the voltage in range of 3.3 to 4.2 volts upto 5 volts which we can provide to the ESP32 through the VIN pin which has a linear voltage regulator attached to it with a fixed voltage output of 3.3 volts.

The new circuit diagram created is shown in Figure 4.6 along with the second prototype robot in Figure 4.7.

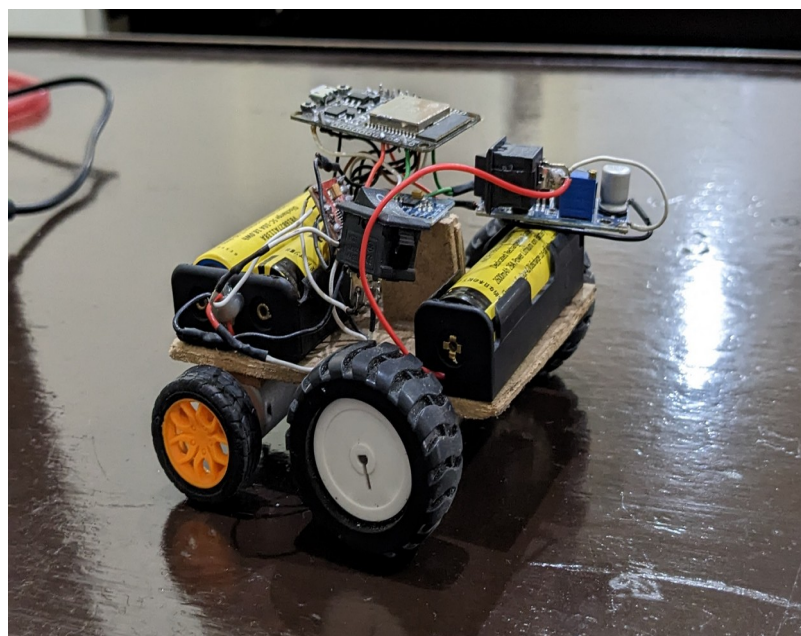
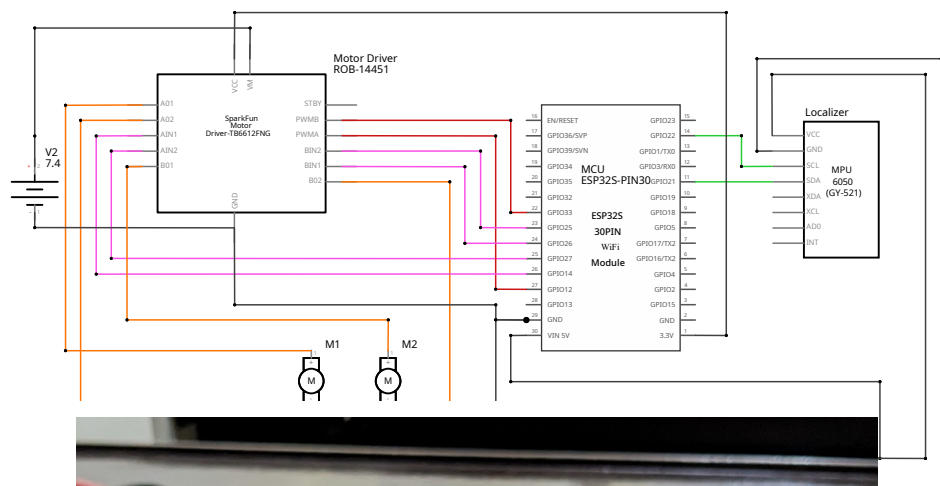


Figure 4.6: The second robot prototype

4.5 The issue with MPU6050

This is where we actually fell onto the wrong track as MPU6050 is not supposed to be a object localizer. Instead it is used along with better localization protocols like GPS to find the orientation of the object which is which way the object is facing and which side it is going towards. It is never the main sensor of the project. The reason is its high error rating with a diversion of about 6 meters per minute. This reading is even greater than the length of our playing field which is 4 foot. This leads us to the third prototype.

4.6 The third prototype of the robot

The third prototype is the current prototype of the project. It is an image processing based approach while keeping the resource usage quite minimal. The mpu6050 was removed in this approach and a mobile camera took its place as the localizer. An IP camera is what was needed but due to its unavailability and a good FOV of an android phone camera, we used used that instead.

The architecture of this prototype is simple and is explained quite well in Figure 3.3. A human setups the camera so that it catches a bird eye view live feed of the field with the robots and the ball. The camera transfers the live feed through high speed wired connection to the laptop where it is processed through a Python script with the help of opencv. The coordinates of each of the objects is extracted from the video feed and is then sent to the robots to act upon them.

The robots have a colored roof added on top of them to facilitate the image processing algorithm with better detection. The robots also have two more square objects on top of them in order to help the algorithm find the orientation of the robot.

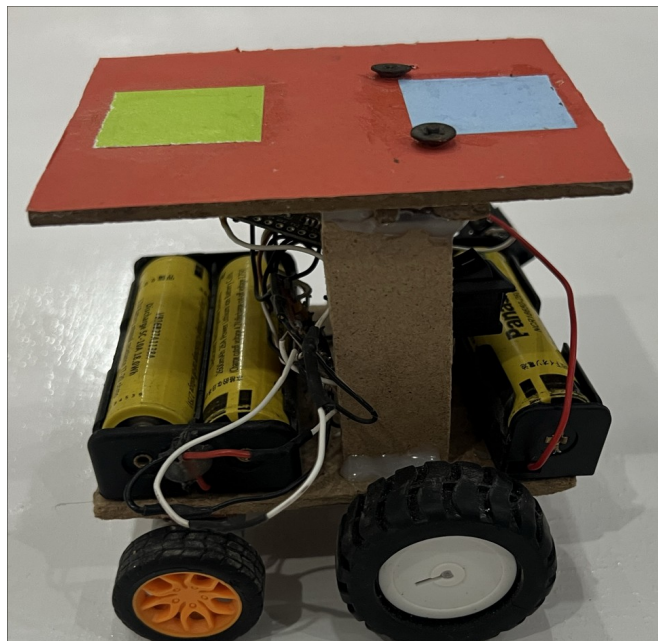


Figure 4.7: The third prototype of the robot

The python code gets the live feed from the android camera and extract the current frame of the video from it using the opencv's `stream.read()` function. The frame is then converted into an hsv image using the opencv's `cvtColor(frame, cv2.COLOR_BGR2HSV)` function. Then

every object is is searched arround in the image with pre-taken hsv values that are comapred to the hsv image using the opencv's `inRange ()` function. This creates binary masks for every single object. The binary masks are then searched for contours of the objects using the opencv's `findContours()` function. The bounding boxes of each object is then calculated using opencv's `boundingRect()` function and then the center coordinates of each object is calculated. The results are then sent to the robot for its movement.

4.7 Prototype 3.1 of the robot

This prototype is named 3.1 because it has the same circuit as of the third prototype. The thing which was changed in this prototype is the way the robots moves after detecting the ball. This is explained in more detail in the Results chapter.

Chapter Five

5 RESULTS

5.1 Overview

Our project aimed to develop a real-time planning and execution system for robots, the main focus being on effective communication, precise localization and navigation. The journey of our project involved several iterations and prototypes, leading to significant findings and outcomes.

5.2 The first prototype

The first prototype of the robot was displayed during the first progress presentation of the project and consisted of a sender module and a receiver module. The actual robot (Figure 4.2) with which the MPU6050 was attached acted as the sender while another temporary circuit with an ESP32 and a attached LCD acted as the receiver. The goal of this prototype was to get readings from the Robot and send it to the receiver using the ESP-NOW protocol. The results we satisfactory as shown in Figure 5.1.

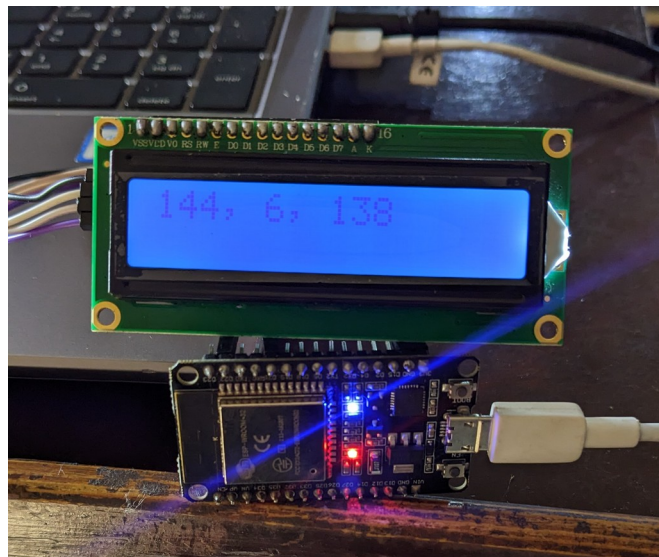


Figure 5.1: Readings recieved from the sender module displayed on the LCD

5.3 The second prototype

The main improvements in the second prototype were the inclusion of a 5V voltage booster to provide the robots with the correct voltage and the change to gear motors before which the car was unable to move. The second prototype robot also featured a different motor driver than before. All of this resulted in a more stable circuit design together with a movable robot and the new motor driver decreasing the voltage drop massively resulting in better battery life.

5.4 The third prototype

The third prototype featured a completely different model for the project. The object detection was shifted towards an image processing based solution instead of a hardware based solution. The hardware also saw some changes like changing the voltage booster from a potentiometer based output to a fixed 5v output. The prototype was demonstrated during the final presentation of the project. The results were successful object detection and successful navigation of the robot to the ball. But there was a issue that the robot will find path to the ball only if the angle between the ball and the robot was very small like about 25-50 degrees. If the angle between the ball and the robot was greater than that, the robot will start to move in circles around the ball and the radius of the circle will never decrease causing the robot to never get to the ball. The results are shown in the images down below

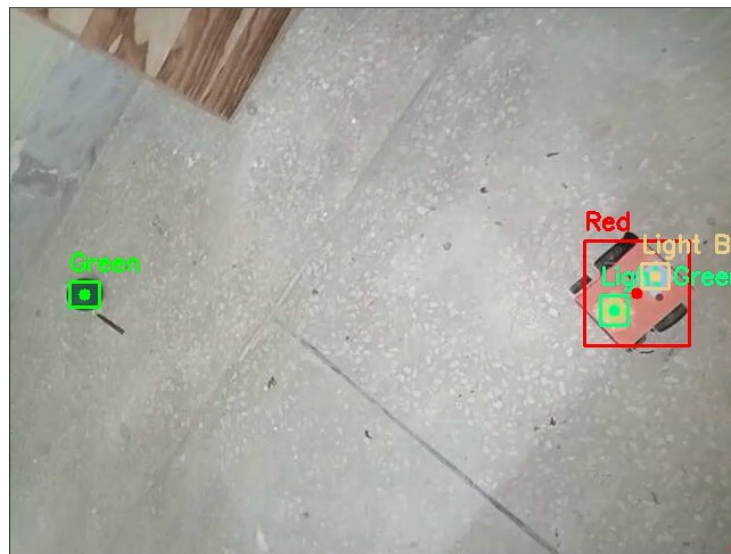


Figure 5.2: The image processing algorithm correctly identifying the ball and robot

```
Relative Angle: 37.97638746084243  
Sent code: 2, bytearray(b'\x02\x00\x00\x00')
```

Figure 5.3: Laptop finding the angle of the ball with respect to the robot and sending the readings to the robot



Figure 5.4: The robot turning towards the ball

```
Relative Angle: 8.248899251591524
Sent code: 0, bytearray(b'\x00\x00\x00\x00')
```

Figure 5.5: Angle between the ball and the robot changing as the robot turns



Figure 5.6: The robot reaching the ball

5.5 Prototype version 3.1

This prototype was demonstrated during the OpenHouse hosted in HITEC University. The main change in this version was the complete change in the behaviour of the how the robot moved towards the ball. Before this once the ball was detected, if the relative angle was less than zero, the robot moved left by making an arc as shown in Figure 5.7:

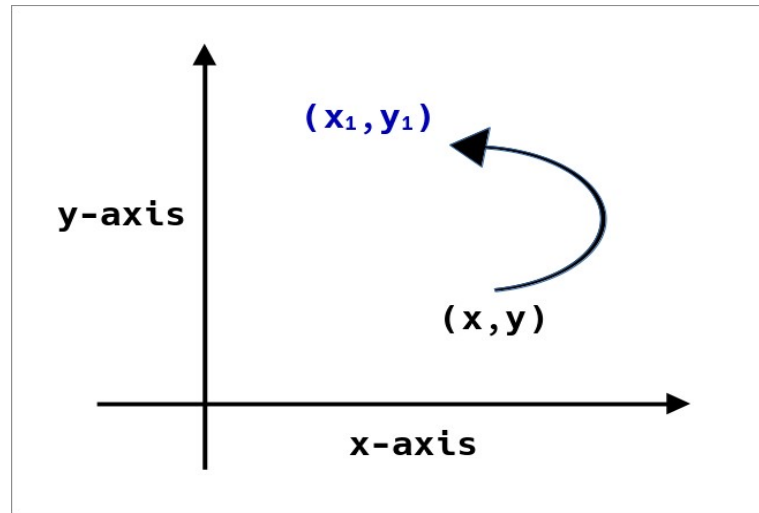


Figure 5.7: Robot movement from its position (x, y) to the ball's position (x_1, y_1)

This movement was changed in version 3.1 so that the robot after detecting the ball first moves backwards in an arc instead of going forward and when the robot's angle with the ball becomes closer to zero degrees, it starts to move forward. This drastically improved the efficiency of our project and made the robot able to get to the ball irrespective of how large the angle is instead of the 25-50 degrees limit in version 3. The results were as follows:



Figure 5.8: Laptop detecting the ball and the robot

```
Sent code: 1, bytearray(b'\x01\x00\x00\x00')  
Relative Angle: -149.76384199391177
```

Figure 5.9: Laptop communicating to the robot to move left

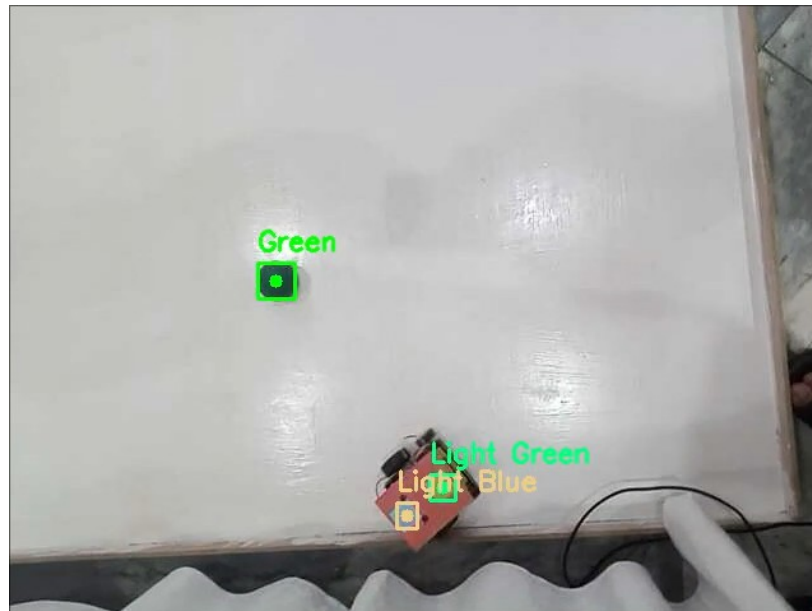


Figure 5.10: Robot reducing the angle between itself and the ball

```
Relative Angle: -78.39191351555169  
Sent code: 1, bytearray(b'\x01\x00\x00\x00')
```

Figure 5.11: The angle reduced from -149 to -78

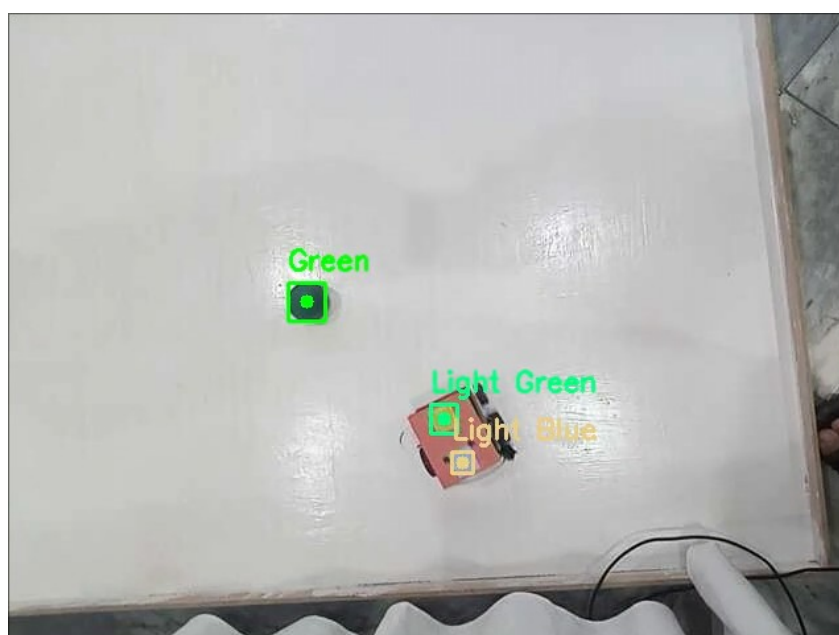


Figure 5.12: The robot almost facing the ball



Figure 5.13: The robot gets to the ball

5.6 Overall Outcomes

The project successfully achieved its primary objectives of precise robot localization and effective real-time communication. The second objective of the realtime navigation turned out to be much more challenging than what we initially thought but in the end it was achieved too by the robot successfully pathfinding to the ball in prototype version 3 and 3.1. We were unable to have more than one robot on the field at a time unfortunately due to time constraints and also due to complete changes in the working methodology from one prototype to the other. The transition from a sensor-based to an image processing-based localization approach significantly enhanced the reliability and accuracy of the system. The use of color segmentation and real-time video processing enabled efficient tracking and coordination of robots, demonstrating the potential for improved performance of robots in industrial and dynamic settings. But as the project was a reasearch project with a main focus on improving our knowledge and contributing to advancements in industrial automation and innovation it aligns with the 9th goal of the United Nations Sustainable Development Goals (SDGs), promoting industry, innovation, and infrastructure.

Chapter Six

6 CONCLUSION

6.1 Summary of the project

The project's aim was to design and implement a real-time planning and execution system for robots, specifically focusing on robotic soccer as it offers a dynamic and practical testing ground. The project journey consisted of multiple iterations and prototypes.

The original approach used IMUs for localization and the calculation of robot's acceleration, velocity, and orientation. But, the sensor noise resulted in high error rate and drift in position by time. This caused a change of methodology.

Due to the shortcomings in the IMU-based localization, the project shifted to an image processing approach using a mobile camera to capture a live video feed of the playing field and video processing using OpenCV. This significantly improved the accuracy and reliability of the localization system as now we were able to identify different objects based on their color and find their location in the video by finding their coordinates. The result was a successful navigation of the robot towards the ball.

6.2 Implications and Limitations

Initially we tried to do a completely hardware based project but were unable to continue that and had to shift to video processing. Video processing is easy, is reliable and gets the job done but video processing can be slow and is really expensive. The video stream we got from the mobile camera was 640x480 pixels and it worked for our project but might not work for a larger scale project. Increasing the resolution will result in the need of a GPU to process the video stream instead of our CPU based model. We were also unable to implement a robot game between two or more robots due to time constraints.

6.3 Recommendations for Future Research

Future work should first consider implementing the actual soccer game between multiple robots and then try to scale the project for larger environments by implementing an algorithm for higher resolution video processing using a GPU and may be even look into making a complete hardware based prototype in order to improve the speed and efficiency of the project even more.

6.4 Final Thoughts

In conclusion, this study was about studying the behaviour of wireless electronic robots in a dynamic environment with multiple changes in the fundamental parts of our robots along the way. In the end, we have a complete working prototype of a robot that can navigate to a specified target with the help of image processing algorithms. We hope that the project will improve with time to have better efficiency of target scoring with greater execution speed.

REFERENCES

- [1] Kitano, Hiroaki & Asada, Minoru & Kuniyoshi, Yasuo & Noda, Itsuki & Osawa, Eiichi., “RoboCup: the Robot World Cup Initiative.” Proceedings of the International Conference on Autonomous Agents, 1996
- [2] Ubbo Visser and Hans-Dieter Burkhard, “RoboCup: 10 Years of Achievements and Future Challenges”, AI Magazine Volume 28 Number 2, 2007